A Clinical Study on Islanders to Determine the Potential Impacts of Music Genres on Running Speed and Heart Rate.

Victoria Chan, Logan Umaguing,
Aly Yang, & Victoria Yee

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1 Abstract

Music and exercise intensity are known to influence physical performance and physiological response. This study investigates their combined effects using a two-way factorial randomized design. Eighty-one participants were assigned to one of nine groups based on music genre (Classical, Heavy Metal, No Music) and run distance (100m, 1km, 5km). Two-way ANOVA results revealed a significant interaction between music and intensity for performance, with heavy metal improving endurance at longer distances. No significant effect was found for heart rate. These findings suggest that music may enhance performance in intensity-specific contexts.

2 Introduction

Music and exercise intensity are two established factors that influence physical performance and physiological response. Music can regulate psychological states and improve exercise outcomes, while intensity affects energy demand and cardiovascular load. Although often studied separately, few studies have examined their combined effects on performance and heart rate.

Previous research highlights the role of music in pre-exercise physiological readiness. For instance, preferred warm-up music has been shown to increase average heart rate compared to no music, suggesting elevated physiological arousal (Karow, 2020). Similarly, listening to fast-tempo music during warm-ups improves movement speed and readiness for high-intensity exercise, leading to enhanced physical performance (Patania et al., 2020).

Beyond warm-ups, music can influence emotional and cognitive functioning during exercise. Music helps individuals regulate emotions, maintain focus, and sustain performance under physical stress (Lane et al., 2011). Fast-tempo music, in particular, has been linked to significantly faster self-paced running speeds and higher peak heart rates (Lee & Kimmerly, 2016). Additionally, music has been shown to modulate arousal, distract from fatigue, and elevate affective states, all of which contribute to improved endurance and perceived exertion (Smoll & Schultz, 1978).

Despite this evidence, the interaction between music genre and exercise intensity has not been thoroughly explored. It remains unclear whether the benefits of music—especially high-arousal genres like heavy metal—are amplified or diminished under different intensity conditions. Understanding this interaction could inform targeted strategies in sports training, rehabilitation, and performance optimization.

Therefore, our study examines how music genre and exercise intensity interact to affect completion time and pulse rate. We hypothesize that music's impact will be strongest during longer runs, while heart rate will be more influenced by intensity than genre.

3 Methods

3.1 Participants

The participants are from all 3 Islands (Ironbard, Providence, and Bonne Santé). A complete list of all Islanders was collected via web scraping R techniques, and then men aged 30-39 were picked as our target population. The seed was set as 123. We then used the sample function to randomly select participants from every island, and to sort them into one of the nine combination groups.

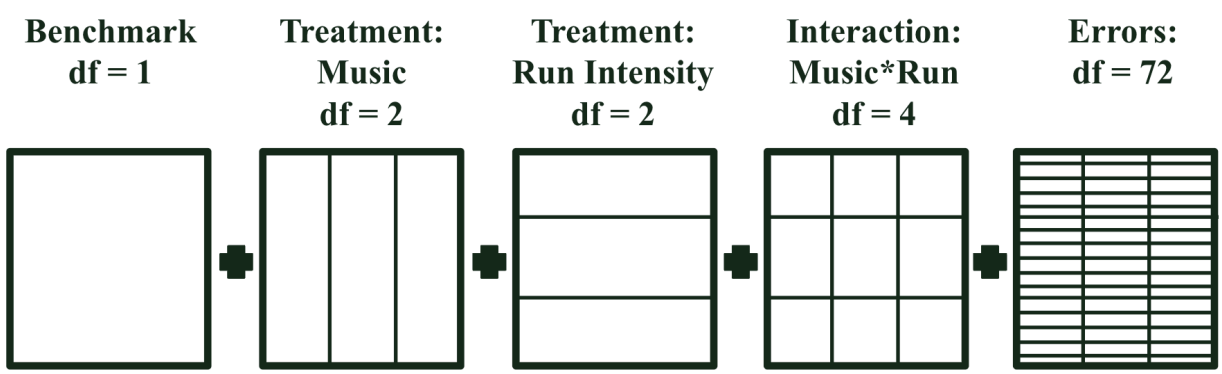
3.2 Design

The study uses a Randomized Basic Factorial Design with Two Factors, and is single blind. The measured parameters are as follows:

Response Variables	Running Time and Heart Rate (BPM)		
Factor 1 (Music)	Control (No Music)	Heavy Metal	Classical
Factor 2 (Run Intensity)	100m	1km	5km

Table 1. Parameters

The factor diagram for degrees of freedom is shown here:



We chose to focus on Heavy Metal and Classical music specifically since they are on opposite ends of the spectrum in terms of music energy, with the former being faster tempo and the latter being slower. We would like to discover whether any factor has an impact on the response variables, and if the interaction between them would be statistically significant. There is a focus on the intensity of each run, as this shows if the type of music has any impact on a long term basis.

3.3 Instruments

Heart Rate was measured both before and after the following steps, from the Islanders using a pulse meter. The two different types of music were played for 10 minutes per person (or not at all for the Control Factor), and then an outdoor run was immediately started. Running time was automatically calculated after completing the run, and recorded using Google Sheets. All recorded information was imported into R Studio for further analysis.

3.4 Procedure

Step 1: Find 81 consenting male participants from the Islands aged 30-39 that are willing to participate in our experiment. Then, we used R to generate a random sample of these participants.

Step 2: Using R code, "randomly" assign these participants into one of the nine different groups for music and run intensity. The groups are as follows:

1. No music, 100m outdoor run
2. No music, 1km outdoor run
3. No music, 5km outdoor run
4. Heavy Metal music, 100m outdoor run
5. Heavy Metal music, 1km outdoor run
6. Heavy Metal music, 5km outdoor run
7. Classical music, 100m outdoor run
8. Classical music, 1km outdoor run
9. Classical music, 5km outdoor run

Step 3: For each Islander, measure their starting Heart Rate (beats per minute) using the pulse meter.

Step 4: Apply the factors listed according to what group the islander is in, first by listening to Heavy Metal Music, Classical, or no music for 10 minutes.

Step 5: Directly after listening, start the designated outdoor run. Record their final run time.

Step 6: Measure their heart rate again. Record the data into the Google Sheet.

Step 7: For each Islander, compare the difference in heart rate (after - before), and record the total time spent running as a response variable.

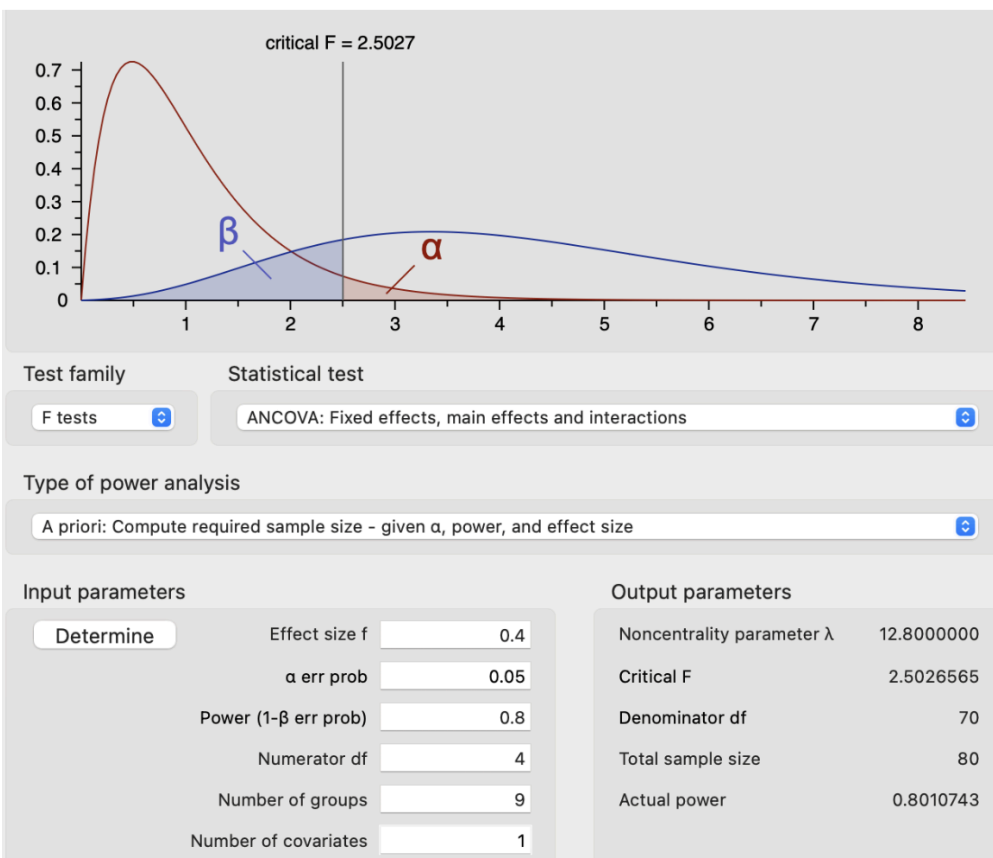
4 Data Analysis

4.1 Type of Statistical Analysis

Using R, we conducted two analyses of variance (ANOVA) on the collected data, once for Heart Rate and once for Running Time. The statistical analysis we used to test if the type of music influences an individual's performance in a high-intensity exercise using variance analysis and post-hoc Tukey HSD, if the results are significant. Using

4.2 Sample Size Determination

We used GPower to determine our total sample size. We chose a power of 0.8 to show that the probability we correctly reject the null hypothesis is 80%. An alpha value of 0.05 was chosen, to minimize the chance of incorrectly rejecting the null hypothesis when it is in fact, true. The numerator degrees of freedom was chosen to be 4, as one degree is added for interaction. We are using a two factor random basic factorial design, and determined that the sample size needed would be 80. To formulate 9 balanced groups, we rounded up to 81 participants.



5 Results - Running Performance

5.1 ANOVA Analysis

Factor	DF	Sum Sq	Mean Sq	F-value	P-value
Music	2	0.10	0.05	0.652	0.5239
Intensity	2	236.19	118.10	938.959	<2e-16
Music:Intensity	4	1.33	0.33	2.651	0.040
Residuals	72	9.06	0.13		

Table 2. Two-Way ANOVA with Blocking and Interaction

The ANOVA results indicate that music genre by itself (p-value = 0.5239, which is greater than 0.05) does not significantly alter running time. By contrast, intensity yields a very significant p-value of <2e-16, suggesting that only the varying intensity levels had a statistically significant effect on the change in running time. Furthermore, the Music x Intensity interaction (p-value = 0.040) remains significant, indicating a combined effect between music genre and intensity level on running time.

5.2 Tukey HSD

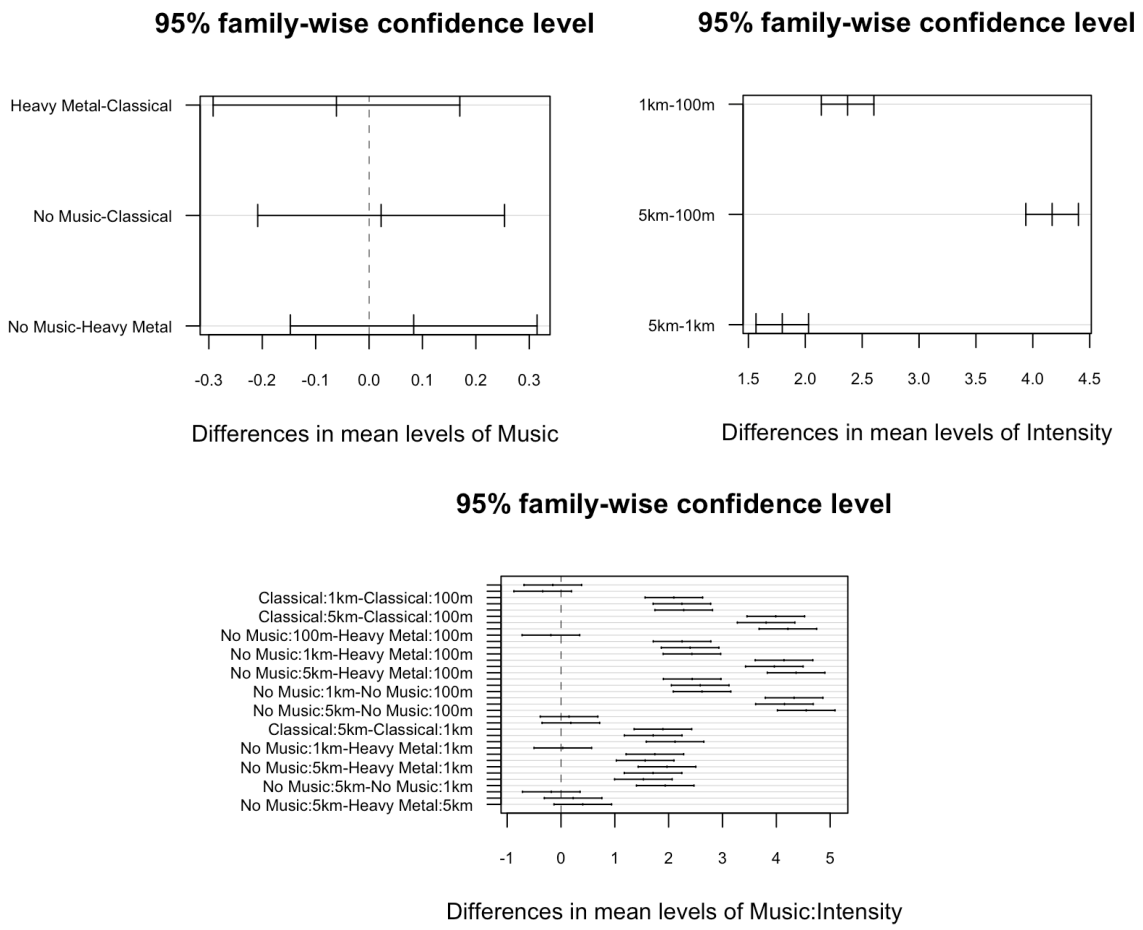


Figure 1. Post-Hoc Analysis of Difference in Change of Running Time between Different Treatment Groups

Utilizing a Tukey HSD procedure to maintain a 95 percent family-wise confidence level, we tested the pairwise contrasts for music, intensity, and their interaction. For music, the confidence interval for heavy metal versus classical, no music versus classical, and no music versus heavy metal all include zero, indicating no significant difference in mean running time among these genres. For intensity, all three comparisons have confidence intervals entirely above 0, indicating that the difference in running time between the running intensity comparisons is significant. Finally, the different Music x Intensity pairings revealed significant differences for many pairings, and a common trend was that as long as the pairings did not have the same intensity, then the comparisons were shown to be significant.

5.3 Residual Diagnostics

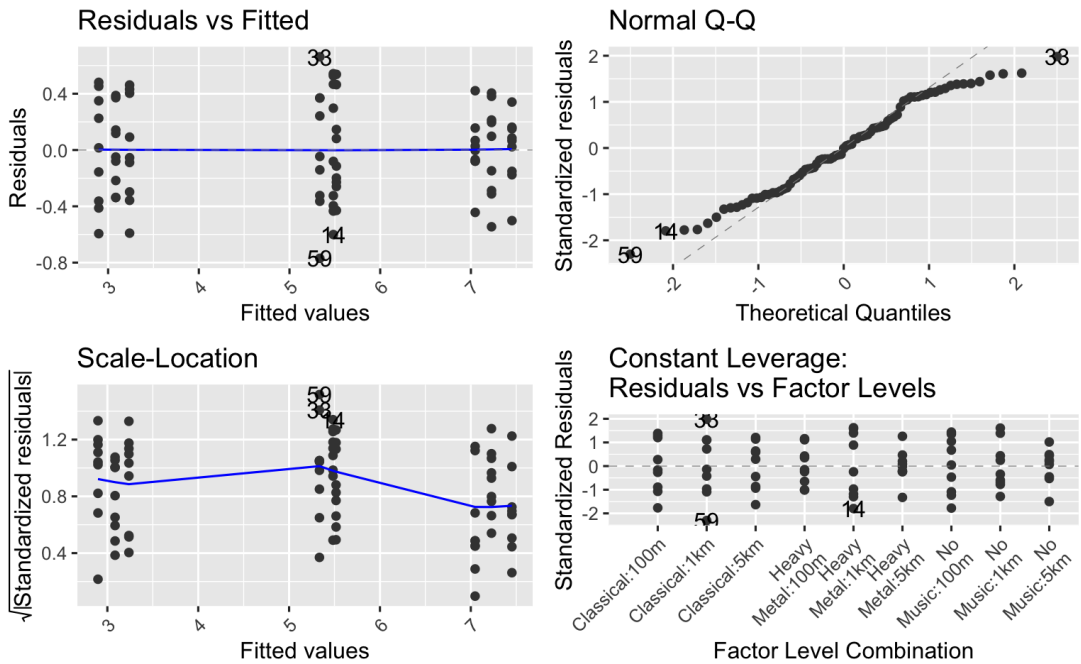


Figure 2. Summary Plots of Residuals for ANOVA Results

Visualized through the deviating tails in a Normal Q-Q plot, the assumption of normally distributed residuals failed to be upheld and suggested that the model required transformation, even though the other assumptions were reasonably satisfied. We explored a Box-Cox transformation and found that the log transformation was ideal and primarily resolved the key issue of non-normality of residuals, although the tails still deviate slightly. Figure 2 shows the residual plots after applying a log-transformation.

5.4 Box Plots

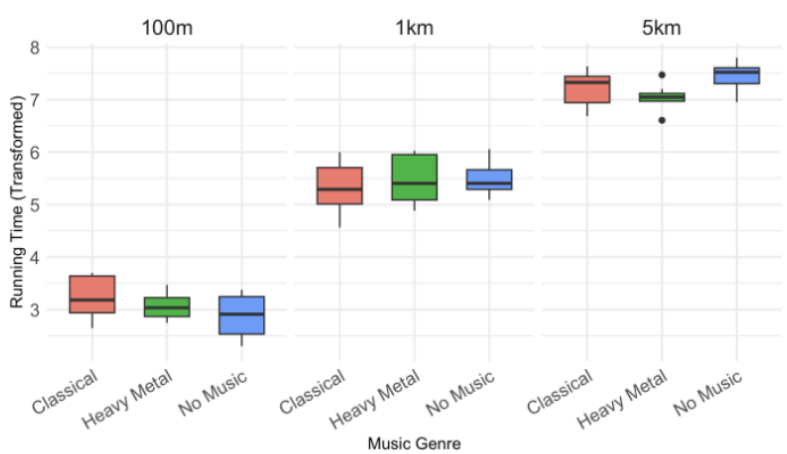


Figure 3. Box Plots Comparing Music Genre and Transformed Running Time

In each panel, the central black line marks the median running time, while the top and bottom of the colored box correspond to the third and first quartiles, respectively. The median has no significant difference between the genres aside from the 5km section. The whiskers extend to the most extreme observations within 1.5× the interquartile range, and any points beyond that threshold are plotted individually as black dots, indicating potential outliers. The outliers within these box plots are exclusive to the Heavy Metal listeners who ran 5km.

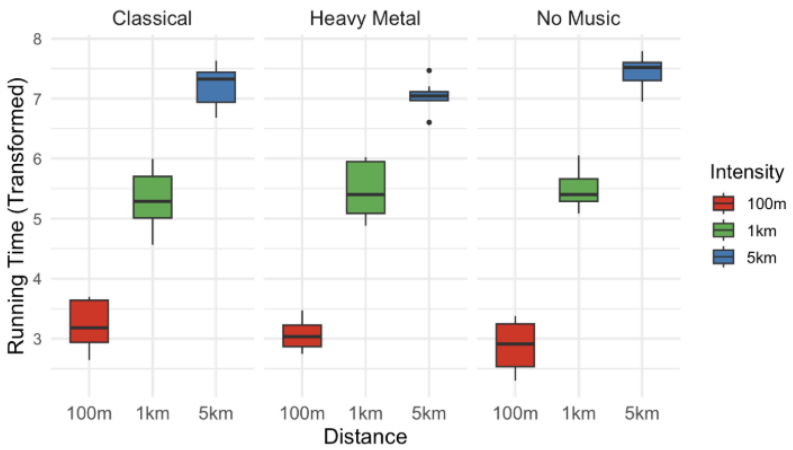


Figure 4. Box Plots Comparing Distance and Transformed Running Time

The box plots in Figure 4 reaffirm the presence of possible significant interactions between music and intensity, as running times noticeably vary for different combinations of distance and music genre.

6 Results - Heart Rate

6.1 ANOVA Analysis

Factor	DF	Sum Sq	Mean Sq	F-value	P-value
Music	2	279.0	139.494	2.1167	0.1279
Intensity	2	464.8	232.383	3.5262	0.0346
Music:Intensity	4	183.2	45.809	0.6951	0.5978
Residuals	72	4744.9	65.901		

Table 3. Two-Way ANOVA with Blocking and Interaction

The ANOVA results indicate that music genre by itself (p-value = 0.1279, which is greater than 0.05) does not significantly alter heart rate. By contrast, intensity yields a p-value of 0.0346, indicating that intensity level does have a statistically significant effect on the change in heart rate. Lastly, the Music × Intensity interaction (p-value = 0.5978) remains non-significant, suggesting there is no evidence of a combined effect beyond the main effect of intensity alone.

6.2 Tukey HSD

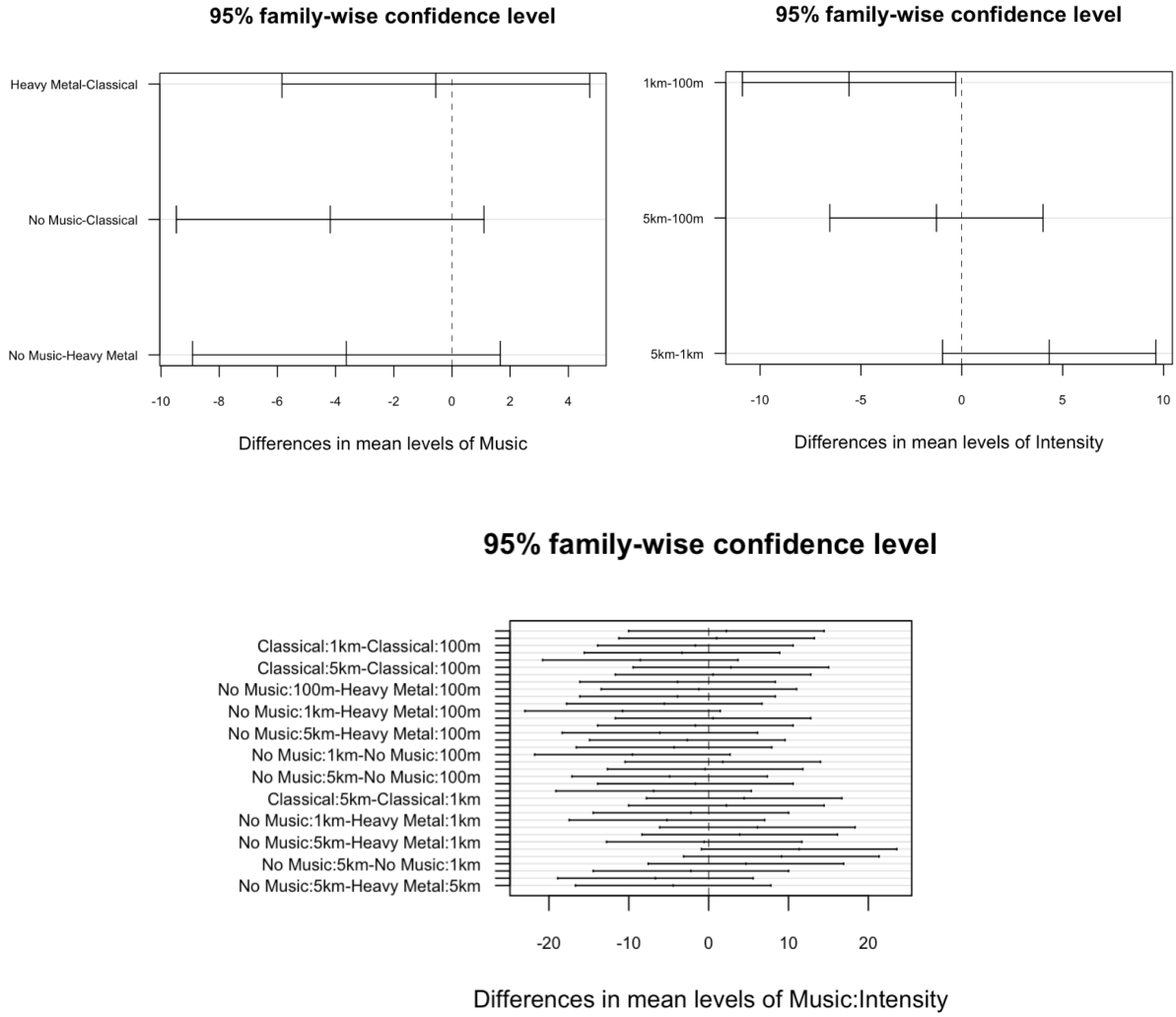


Figure 5. Post-Hoc Analysis of Difference in Change of Heart Rate between Different Treatment Groups

Using Tukey's Honest Significant Difference procedure to maintain a 95 percent family-wise confidence level, we tested all pairwise contrasts for music, intensity, and their interaction. For music, the confidence intervals for heavy metal versus classical, no music versus classical, and no music versus heavy metal all include zero, indicating no significant differences in mean heart rate change among these genres. For intensity, the interval for 1km versus 100m excludes zero, revealing a significant decrease in heart rate when the distance increases from 100m to 1km. In contrast, the intervals for 5km versus 100m and 5km versus 1km include zero, showing no evidence of a difference in heart rate change at longer distances. Finally, across every music by intensity pairing, all intervals include zero, demonstrating no significant interaction effects once we correct for multiple testing.

6.3 Residual Diagnostics

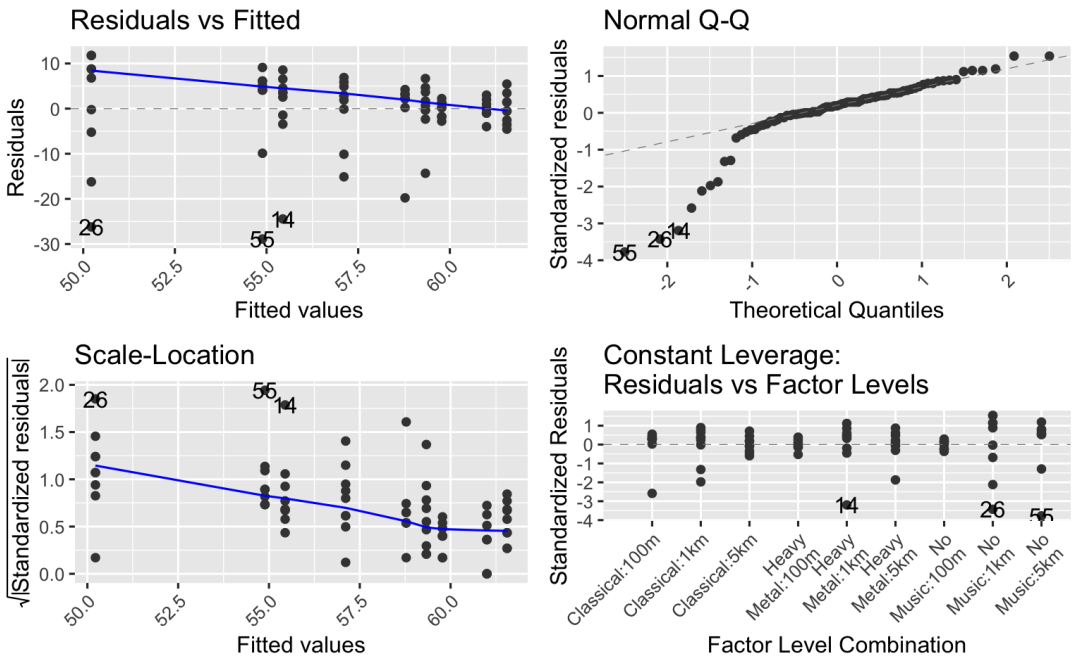


Figure 6. Summary Plots of Residuals for ANOVA Results

We initially explored a Box-Cox transformation, which yielded an estimated maximum likelihood value of $\lambda = 2$ but failed to resolve key issues such as non-normality and non-constant variance. As a result, we retained the original model without transforming the response variable.

6.4 Box Plots

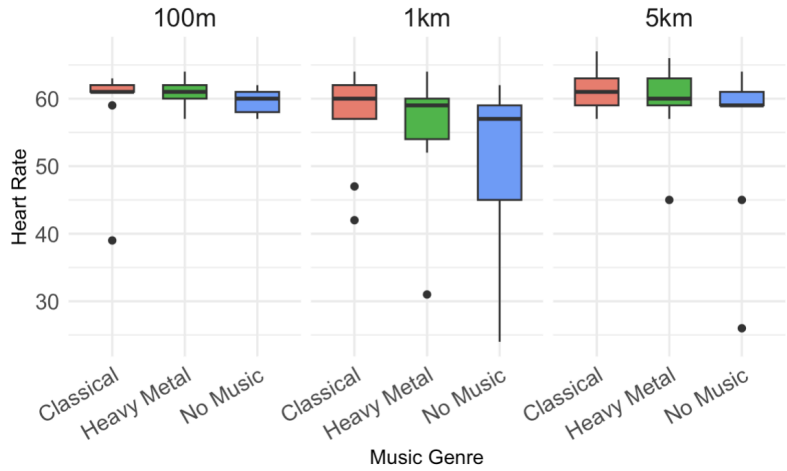


Figure 7. Box Plots Comparing Change in Heart Rate between Music Genres.

In each panel, the central black line marks the median heart-rate change, while the top and bottom of the colored box correspond to the third and first quartiles, respectively. The whiskers extend to the most extreme observations within 1.5× the interquartile range, and any points beyond that threshold are plotted individually as black dots, indicating potential outliers.

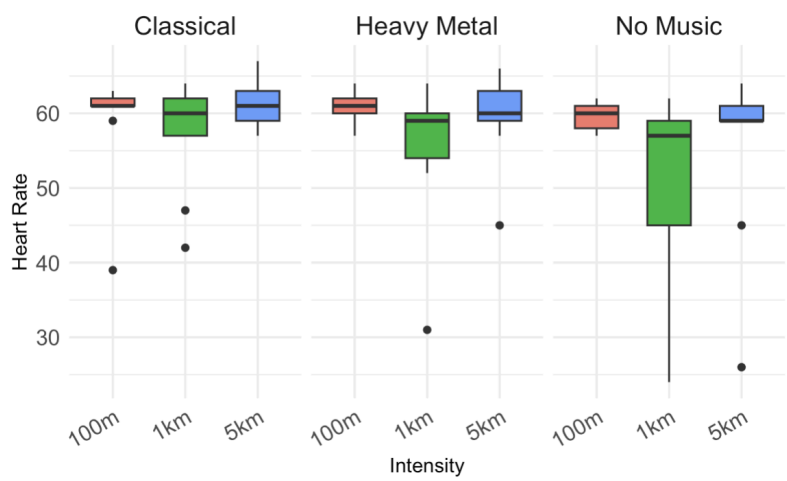


Figure 8. Box Plots Comparing Change in Heart Rate between distance.

In each panel, the central black line marks the median heart-rate change, while the top and bottom of the colored box correspond to the third and first quartiles, respectively. The whiskers extend to the most extreme observations within 1.5× the interquartile range, and any points beyond that threshold are plotted individually as black dots, indicating potential outliers.

7 Discussion

The aim of our study was to evaluate how different music genres and exercise intensities affect running performance and heart‑rate responses. We first imported the full dataset into R and fit two separate two‑way linear models—one for running performance and one for change in heart rate, each including music genre, run intensity, and their interaction. For both models, we inspected residual diagnostic plots to verify the model's adequacy. We log-transformed the response variable for running performance (due to the normality assumptions being violated) while keeping the heart rate response variable in its original scale to satisfy model assumptions.

Going deeper into our analysis, we found that both exercise intensity and the Music × Intensity interaction significantly affect running performance, while only exercise intensity has a statistically significant effect on heart rate. The Post-Hoc Tukey HSD test was used to perform the pairwise comparisons and reaffirmed that there was no significant difference in running time for the music genres, but there was a significant difference between all running intensities. This suggests that the higher the intensity, the higher the running time, which was already expected. Furthermore, the Tukey HSD tests for the Music x Intensity suggested that there was a significant interaction between certain pairings as long as the pairings did not have the same intensity. This suggests that the interaction between music and intensity is primarily dependent on the differences in intensity. The Post-Hoc Tukey HSD test for the heart rate suggests there is no significant difference in mean heart rate among the different music genres, and only showed a significant difference between running intensities for the 1km versus 100m pairing. This suggests that heart rate becomes relatively consistent at longer distances, but rises significantly for 100m. Across the Music x Intensity pairings, there were no significant interaction effects for heart rate.

The box plots provide valuable insight into how running performance and heart rate vary across different music genres and exercise intensities. In Figures 3 and 4, we observe that as running distance increases, the variability in transformed time also grows, most notably in the 5km trials. Participants who ran without music generally had longer times at longer distances, suggesting that music may provide a motivational advantage during more strenuous runs. Participants who listened to heavy metal tended to have shorter running times in the 5km trial, indicating it may be particularly effective in enhancing endurance performance. In Figures 7 and 8, there is no consistent trend across Classical, Heavy Metal, or No Music conditions, as their median heart rates are very similar. This supports the ANOVA result showing that music genre does not have a significant effect on heart rate.

We recognize several limitations in our factorial design that should be addressed in future research. First, conducting all trials outdoors introduced uncontrolled environmental variability. Factors such as temperature, wind, and surface conditions may have confounded both speed and heart rate measurements. Using an indoor environment might help reduce this source of noise.

Second, we observed noticeable differences in running performance among participants from different islands. This suggests that geographical or environmental background may systematically influence outcomes. Including island as a blocking factor in the design would allow us to control for these baseline differences and better isolate the effects of music genre and exercise intensity on heart rate and running performance. Future experiments should consider stratifying by island to reduce between-group variability and enhance the interpretability of treatment effects.

In conclusion, our study found that music and the intensity of the run do have statistical interaction based on the tested models, but did not find a significant effect when music became a factor.

8 References

Karow, M. C., Rogers, R. R., Pederson, J. A., Williams, T. D., Marshall, M. R., & Ballmann, C. G. (2020). Effects of Preferred and Nonpreferred Warm-Up Music on Exercise Performance. *Perceptual and Motor Skills*, *127*(5), 912-924. https://doi.org/10.1177/0031512520928244 (Original work published 2020)

Lane, Andrew M., Davis Paul A., and Devonport Tracey J. (2011). "Effects of Music Interventions on Emotional States and Running Performance". *The Journal of Sports Science and Medicine,* 10: 400–407.

Lee, Sam, and Derek S Kimmerly. "Influence of music on maximal self-paced running performance and passive post-exercise recovery rate." *The Journal of sports medicine and physical fitness* vol. 56,1-2 (2016): 39-48.

Patania, Vittoria Maria, et al. "The Psychophysiological Effects of Different Tempo Music on Endurance versus High-Intensity Performances." *Frontiers in Psychology*, vol. 11, no. 74, 5 Feb. 2020, www.ncbi.nlm.nih.gov/pmc/articles/PMC7013107/, https://doi.org/10.3389/fpsyg.2020.00074.

Smoll, F L, and R W Schutz. "Relationships among measures of preferred tempo and motor rhythm." *Perceptual and motor skills* vol. 46,3 Pt 1 (1978): 883-94. doi:10.2466/pms.1978.46.3.883